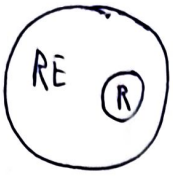
Lab 5 - AA



$f \leq_m g$

$f \notin R \Rightarrow g \notin R$

$RE \Rightarrow RE$

$g \in R \Rightarrow f \in R$

$RE \quad RE$

1. ex de MT pt. care nu putem să det. algoritmic că se termină.

   $HP \in RE \setminus R$

   $MTU(M, w)$

2. ex. de MT pt. care putem det. algoritmic că se termină: ~~Ungarian~~

   temă cu sudoku

   nr. par sau impar

3. a) $f_{or}(M_1, M_2) =$ { TRUE <=> $M_1[111] \rightarrow$ TRUE V $M_2[111] \rightarrow$ TRUE ; FALSE, altfel

   $\notin R, \in RE$

$M_{or}(M_1, M_2)$:

```
while(1):
    step(M1[111])
    step(M2[111])
    if(M1[111] → TRUE sau M2[111] → TRUE)
        return 1;
```

$f_H \notin R$

$f_L \notin RE$

$f_H \leq_m f_{or}$

$M, w \xrightarrow{t} M_1, M_2$

$M_1(x)$: return FALSE

$M_2(x)$: run $M(w)$

return TRUE

„⇒" $f_H(M, w) = 1 \Rightarrow M[w]$ s-a terminat $\Rightarrow M_2[w] \to TRUE$, $\forall w$

$\Rightarrow M_2[111] \to TRUE \Rightarrow f_{or}(M_1, M_2) = 1$

„⇐" $f_{or}(M_1, M_2) = 1 \Rightarrow M_2[111] \to TRUE \Rightarrow M[w]$ s-a terminat

$\Rightarrow f_H(M, w) = 1$

$\Rightarrow f_H \leq_m f_{or} \;|\Rightarrow f_{or} \notin R$

$\hookrightarrow \notin R$

c) $f_{diff}(M_1, M_2) = \begin{cases} TRUE \iff M_1[111] \to TRUE \wedge M_2[111] \not\to TRUE \\ FALSE, \text{ altfel} \end{cases}$

$M_1[111] \to TRUE$ și $M_2[111] \to FALSE$

$M_1[111] \to TRUE$ și $M_2[111]$ ciclează

$\notin RE$

$f_L(M, w) = \begin{cases} TRUE, \text{ dacă } M(w) \text{ ciclează} \\ FALSE, \text{ altfel} \end{cases}$

MT care acceptă: răspunde la TRUE
ciclează la FALSE

$f_L \leq_m f_{diff}$

$M, w \xrightarrow{t} M_1, M_2$

t: $M_1[x]$: return TRUE / tranziție în Y

$M_2[x]$: ~~return~~ run $M[w]$
return TRUE

„$\Rightarrow$" $f_L(M, w) = TRUE \Rightarrow M[w]$ ciclează $\Rightarrow M_1[x]$ ciclează $\Rightarrow M_2[111] \not\to TRUE$

Cum $M_1[111] \to TRUE \Rightarrow f_{diff}(M_1, M_2) = TRUE$

„$\Leftarrow$" $f_{diff}(M_1, M_2) = TRUE \Rightarrow M_2[111] \not\to TRUE \Rightarrow M[w]$ ciclează

$\Rightarrow f_L(M, w) = TRUE$

$\Rightarrow f_L \leq_m f_{diff}$ | $\Rightarrow f_{diff} \notin RE$

$f_L \notin RE$

4. a) $f_S(M) = \begin{cases} TRUE \longleftrightarrow \forall x, M[x] \text{ nu mută niciodată cursorul la stg.} \\ FALSE, \text{ altfel} \end{cases}$

	0	1	#
q_1	$(q_1, 1, \rightarrow)$	$(Y, \#, -)$	$(N, \#, -)$
q_2	$(q_1, 0, \rightarrow)$	$(N, 0, -)$	$(q_1, 1, \rightarrow)$
q_3	$(\ldots, 0, \leftarrow)$	$(\ldots, 1, \leftarrow)$	$(\ldots, \#, \leftarrow)$

$(q_1, w) \vdash^* (q_3, 0) \Rightarrow$ am făcut la stânga

$M_S(M)$: ne uităm în tranzițiile mașinii M

Vom construi graful mașinii M

$(q_1) \xrightarrow{c} (q_2)$ dacă $\delta(q_1, c) = (q_2, ?, X)$

dacă X este $\leftarrow$, punem starea într-o listă

$\triangleright (q_{init})$ găsim toate stările în care putem ajunge din q_{init} indiferent de input

$S \cap L = ? \; \phi \rightarrow TRUE$

$\neq \phi \rightarrow FALSE$

5. $f_{50}(M) = \begin{cases} \text{TRUE dacă } M[1010] \text{ nu trece prin mai mult de 50 tranziții} \\ \text{FALSE, altfel} \end{cases}$

$M_{50}(M)$:
for(i=0; i<50; i++)
step M[1010]
if M a ajuns în Y sau N
return TRUE
else return FALSE

$f_{50b}(M)$ = TRUE dacă $\forall w$, M[w] nu trece prin mai mult de 50 de tranziții

↓

$\$ \notin R$ dem. cu step
$\in RE$

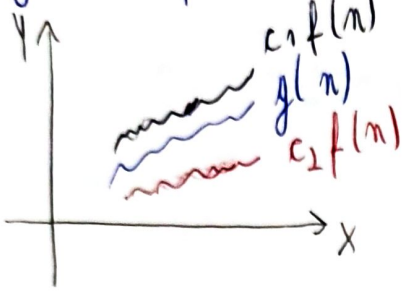
pără NU: $\in R$

Lab 6 - AA

Notații asimptotice

Notația Θ

- $g(n) \in \Theta(f(n))$: $\exists c_1, c_2 \in \mathbb{R}^+$ a.î. $\forall n \geq n_0 \quad c_1 f(n) \leq g(n) \leq c_2 f(n)$



g are aceeași rată de creștere

$f(n) = n^2$
$g(n) \in \Theta(n^2)$ $\Rightarrow g(n) = an^2 + bn + c$

- $g(n) \in O(f(n)) \quad \exists c \in \mathbb{R}^+$ a. î. $\forall n \geq n_0 \quad 0 \leq g(n) \leq f(n) \cdot c$



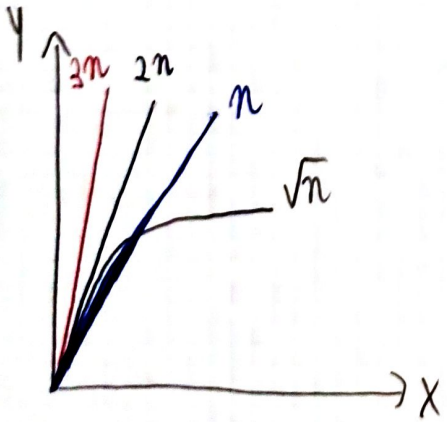
g are rată mai mică sau egală de creștere

```
for(i=1; i<=n; i++)
    if(i % (n-1) == 0)
        break;
```

$g(n) = n-1 \in \Theta(n)$
$\in O(n)$

$n-1$ pași
2 pași
$g(2) \in O(n)$
$g(2) \notin \Theta(n)$

Ex.

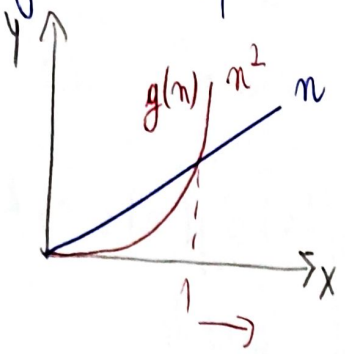


$\Theta(n)$
$g_1(n) = 2n \in \Theta$ și $\in O$
$g_2(n) = \sqrt{n} \in O$
$c_1 n \leq \sqrt{n}, \forall n \geq n_0$
$c_1 \sqrt{n} \leq 1 \Rightarrow \sqrt{n} \leq \frac{1}{c_1}, \forall n \geq n_0$

$\sqrt{n} \to \infty$
$\frac{1}{c_1}$ constant

$\Rightarrow \sqrt{n} \notin \Theta(n)$, dar $\sqrt{n} \in O(n)$

• $g(n) \in \Omega(f(n))$: $\exists c \in \mathbb{R}^+$ a. î. $n \geq n_0$ $\quad cf(n) \leq g(n)$



g are rată de creștere mai mare decât f

Ex. • $f(n) = n$
$g(n) = 3n+1$
• $3n \leq 3n+1$, $\forall\ n \geq 0$
• $f(n) = n$
$g(n) = n^2$
$cn \leq n^2$
$c=1 \Rightarrow n \leq n^2 \quad n \geq 0 \Rightarrow 1 \leq n$

• $g(n) \in \Theta(f(n))$: $\forall c \in \mathbb{R}^+, \forall n \geq n_0 \quad 0 \leq g(n) \leq cf(n)$

Ex. 1) $g(n) = 4n^2+2n \quad g(n) \overset{?}{\in} \Theta(n^2)$ nu

$f(n) = n^2$

$4n^2+2n \leq c \cdot n^2 \quad \forall c \in \mathbb{R}^+, n \geq n_0$

$4n+2 \leq cn$

$2 \leq (c-4)n$

pentru $c=4$

$4n^2+2n \leq 4n^2$

$2n \leq 0$ nu se poate

$\downarrow$

∞

2) $g(n) = \sqrt{n} \overset{?}{\in} \Theta(n^2)$ da

$\sqrt{n} \leq cn^2 \quad \forall c \in \mathbb{R}^+, n \geq n_0$

$0 \leq cn^{\frac{3}{2}} \quad \forall c \in \mathbb{R}^+, n \geq n_0$

- $g(n) \leq \omega(f(n))$ $\forall c \in \mathbb{R}^+, \forall n \geq n_0$ $0 \leq c f(n) \leq g(n)$
  rată de creștere strict mai mare

Exemple de funcții din $O(n)$

$g(n) = const = 2$

$\log_2 n$

$5\sqrt{n}$

$\sin n$

$\sin \in [-1, 1], \forall n$

$0 \leq \sin n \leq c_2 \cdot n$

$|\sin| \leq 1 \leq c_2 \cdot n$

$c_2 = \frac{1}{4}$, $\forall n \geq 4$ egalitatea va fi ok

Exemple de funcții din $\Omega(\log n)$

$\log n \leq \sqrt{n}$

$an + b$

n^μ, $\mu \geq 1$

μ^n, $\mu > 1$

$n!$

n^n

Exemple din $\omega(\frac{1}{n})$

$g(n) = c$ (const)

$g(n) = \sum_{i=1}^{n} a n^i$

$g(n) = \mu^n$, $\mu \geq 1$

$O(f(n) \cdot const) = O(f(n))$

↳ poate fi oricare dintre notații

$\mu^n = 1 \geq \frac{1}{n}$

$\mu = 1$

$\to \Omega(\log n)$ $\quad 1 \in 1^n \stackrel{?}{\in} \Omega(\log n)$

$c \log n \le 1$

$\log_2 n \le \frac{1}{n} \Rightarrow n \le 2^{\frac{1}{n}} = \sqrt[n]{2}$

Dem.: $\log n \in \Theta(\sqrt{n})$

$0 \le \log_2 n \le c\sqrt{n} \mid \cdot 2^n \quad \forall c \in \mathbb{R}^+, n \ge n_0$

pentru $n \le 2^{c\sqrt{n}}$

$1 \le \frac{2^{c\sqrt{n}}}{n}$

$\frac{\log_2 n}{\sqrt{n}} \le c, \forall c \in \mathbb{R}^+$

Dem. $100n^2 \in O(n)$

$0 \le 100n^2 \le 2^n c \quad \forall n \ge n_0$

$100 \le 2^n \cdot c \Rightarrow \forall c \ge 50$ alegem $c = 500$

$100n^2 \le 2^n \cdot 500$

$n^2 \le 2^n \cdot 5$

$2 \le n \log_n 2 + \log_n 5 \quad n \ge 1$

Dem. $10n \in \Theta(2n)$

$c_1 \cdot 2n \le 10n \le c_2 \cdot 2n \Rightarrow c_1 = 1, c_2 = 40$ (cerc mai mare decât 5)

Ex. 1) $f, g \quad f(n) \in O(g^2(n))$

$0 \le f(n) \le c\, g^2(n)$

$g(n) = n \Rightarrow 0 \le f(n) \le cn^2$

$f(n)$ poate fi $n, n^2, \log n$

2. $f(n) \in \omega(\log(g(n)))$

$g(n) = n$

$f(n) \in \omega(\log_2 n) \Rightarrow f(n) = n^2$

3. $f(n) \in \Omega(f(n) \cdot g(n))$ $f(n)$ să crească mai repede sau egal
   - ↳ constant
   - ↳ sau descrescătoare

$\Omega(f(n)) = \Omega(\frac{1}{2} f(n))$

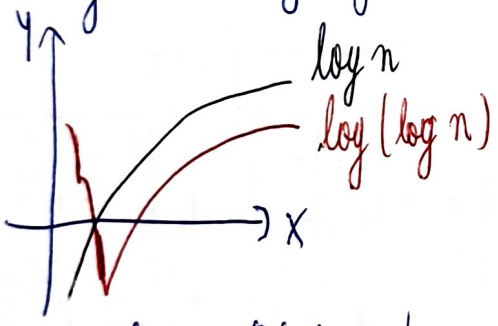
$g(n) = 4 \Rightarrow f(n) \in \Omega(f(n)) \Rightarrow c = \frac{1}{g}, \forall\ 0 \le c \le \frac{1}{4}$

4. verificați valoarea de adevăr

$\sqrt{n} \in O(\log n)$ fals

$\sqrt{n} \ge \log n, \forall n \ge 1$

$\log n \in O(\log(\log n))$



fals

$\forall n \ge 2, \log n \ge \log(\log n)$

$n + \log n \in O(n)$ adevărat

$c_1 n \le n + \log n \le c_2 n$

$\log n \le n \mid + n \Rightarrow c_1 = 1$

$c_2 = 2$

$n \le n + \log n, n \ge 1$

5. $f(n) = \Theta(n) + O(n^2)$ syntactic sugar

$f(n)$: $\exists\, g \in \Theta(n)$ și $\exists\, h \in O(n^2)$ a. î. $f(n) = g(n) + h(n)$

$\Theta(n) \subseteq O(n^2)$ doar așa e validă, invers nu

pt. $\forall f \in \Theta(n), \exists\, g \in O(n^2)$ a. î. $f(n) = g(n)$

$O(n^2) = \Theta(n)$

$\forall f \in O(n^2), \exists\, g \in \Theta(n)$ a. î. $f(n) = g(n)$ $\Big|$ $\Rightarrow$ citirea așa nu e validă

EXERCIȚII EXAMEN

① $\dfrac{O(n\sqrt{n})}{\Theta(n)} = O(\sqrt{n})$ adică ~~toți~~ $\forall f \in O(n\sqrt{n})$ și $\forall g \in \Theta(n)$, $\exists\, h \in O(\sqrt{n})$ a. î. $h(n) = \dfrac{f(n)}{g(n)}$

$0 \le f(n) \le c \cdot n\sqrt{n}$ $\exists\, c_1, c_2, c \in \mathbb{R}^+$

$0 \le c_1 n \le g(n) \le c_2 n$ / inversare $\Rightarrow$

$\Rightarrow \dfrac{1}{c_2 n} \le \dfrac{1}{g(n)} \le \dfrac{1}{c_1 n}$ /·

$0 \le \dfrac{f(n)}{g(n)} \le \dfrac{c}{c_1} \cdot \dfrac{n\sqrt{n}}{n} \Rightarrow \exists\, c_3 \in \mathbb{R}^+$ a. î. $0 \le \dfrac{f(n)}{g(n)} \le c_3\sqrt{n}$

$\dfrac{f(n)}{g(n)} = h(n)$

② Aduceți la forma cea mai simplă

$\dfrac{\Theta(n)}{O(\log n)}$ $\quad c_1 n \le f(n) \le c_1 n$ $\exists c_1, c_2 \in \mathbb{R}^*$

$0 + \le g(n) \le \log n \cdot c$

$0 \le \dfrac{1}{c \log n} \le \dfrac{1}{g(n)} \le \infty$ Ω

$\left(\dfrac{c_1}{c}\right) \cdot \dfrac{n}{\log n} \le \dfrac{f(n)}{g(n)} \le \infty$ Ω(?)

$\exists c_3 = \frac{c_1}{c}$ a.i. $\frac{f(n)}{g(n)} \leq c_3 \cdot \frac{n}{\log n}$ $\quad \Omega\left(\frac{n}{\log n}\right)$

③ ~~$\Theta(n^2)$~~

$\Theta(\sqrt{n})$

$0 \leq f(n) \leq cn^2, \forall c \in \mathbb{R}^+$

$c_1\sqrt{n} \leq g(n) \leq c_2\sqrt{n} \quad \exists c_1, c_2 \in \mathbb{R}^+$

$\frac{1}{c_2\sqrt{n}} \leq \frac{1}{g(n)} \leq \frac{1}{c_1\sqrt{n}} \mid \cdot \Rightarrow 0 \leq \frac{f(n)}{g(n)} \leq \frac{c}{c_1} \cdot \frac{n^2}{\sqrt{n}}$

c_1 fix
c oarecare $\Rightarrow \frac{c}{c_1}$ variază

$\forall \frac{c}{c_1} = c_3 \in \mathbb{R}^+ \quad \Theta\left(\frac{n^2}{\sqrt{n}}\right) = \Theta(n^{\frac{3}{2}})$

④ $f(n) \in \Omega(\log n)$ DA

$g(n) \in O(n)$

$\overset{?}{\Rightarrow} f(n) \in \Omega(\log(g(n)) \quad \exists c_1, c_2, c_3$

$c_1 \log n \leq f(n)$

$0 \leq g(n) \leq c_3 n \mid : c_3$

$0 \leq \frac{g(n)}{c_3} \leq n$

$\log n \geq \log(g(n)) - \log(c_3) \mid \cdot c_1$

$f(n) \geq c_1 \log n \geq c_1 \log(g(n)) - \underbrace{c_1 \log(c_3)}_{\| c_4}$

$f(n) \geq c_1 \log(g(n)) - c_4$

$f(n) \geq c_1 \log(g(n)) - c_4 \geq c_5 \log(g(n)) \quad \text{pt. } n \geq n_0$

Lab 7 - AA

Teorema Master

$T(n) = aT\left(\frac{n}{b}\right) + f(n)$; $a \geq 1, b \geq 1$

I: $f(n) = \theta(n^c)$; $c < \log_b a \Rightarrow T(n) = \theta(n^{\log_b a})$

II: $f(n) = \theta(n^{c \log_b^k})$ $k \geq 0$, $c = \log_b a \Rightarrow$

$\Rightarrow T(n) = \theta(n^{\log_b a} \cdot \log^{k+1} n)$

III: $f(n) = \theta(n^c)$, $c > \log_b a \Rightarrow T(n) = \theta(f(n))$

1.1) $T(n) = 4 \cdot T\left(\frac{n}{4}\right) + 1$

$a = 4$
$b = 4$ $\quad \log_b a = 1$

$f(n) = 1 = \theta(1) \Rightarrow c = 0$

$c < \log_b a$

$0 < 1 \Rightarrow$ Master I $\Rightarrow T(n) = \theta(n^{\log_b a}) = \theta(n)$

1.2) $T(n) = 8 \cdot T\left(\frac{n}{2}\right) + 2n^2$

$a = 8$
$b = 2$
$f(n) = 2n^2 = \theta(n^2) \Rightarrow c = 2$

$c \square \log_b a$ $\quad$ Master I $\Rightarrow T(n) = \theta(n^3)$

$2 \boxed{<} 3$

1.3) $T(n) = 4T\left(\frac{n}{2}\right) + n^2$

$a = 4$

$b = 2$

$f(n) = n^2 \in \Theta(n^2) \Rightarrow c = 2$

$c \boxminus \log_b a$ $\quad n^2 \cdot \log^0 n \Rightarrow \boxed{k = 0}$

$2 \boxminus 2$

Master II $\Rightarrow T(n) = \Theta(n^2 \cdot \log^1 n) = \Theta(n^2 \cdot \log n)$

```
int search(int arr[], int lo, int hi, int v):
T(n)  Θ(1): int mid = (lo + hi) / 2;
      Θ(1): if (arr[mid] == v) return mid;
      Θ(1): if (hi == lo) return -1;
      Θ(1): if (arr[mid] > v)
       T(n/2)  return search(arr, lo, mid, v);
T(n/2)  else return search(arr, mid+1, hi, v);
```

$T(n) = 1 \cdot T\left(\frac{n}{2}\right) + \Theta(1)$

$a = 1$

$b = 2$

$\log_b a = 0$

$f(n) = \text{const} \in \Theta(n^0) \Rightarrow c = 0$

$K = 0$

Metoda arborilor

		const.
niv. 0:	$T(n)$	c
	↓	
niv 1:	$T(n/2)$	c
	↓	
niv 2:	$T(n/4)$	c
	↓	
niv 3:	$T(n/8)$	c
	.....	
niv k:	$T\left(\frac{n}{2^k}\right)$	1

$T(n) =$ $\sum$ costuri pe toate niv. + costul frunzelor =

$= \sum_{i=0}^{k-1} c + \underbrace{\text{nr frunze}}_{1} \cdot 1$ ← "cost frunze" $= k \cdot c + 1 = \log_2 c + 1 = \theta(\log n)$

Metoda substituției

Pp. că $T\left(\frac{n}{2}\right) \in \theta\left(\log\left(\frac{n}{2}\right)\right)$

$\Rightarrow T(n) \in \theta(\log n)$

~~$T(2n) = T(n) + \theta(1)$~~ $\quad T(n) = T\left(\frac{n}{2}\right) + \theta(1)$

$\exists\, c_1, c_2 \in \mathbb{R}^*$ a. î. $c_1 \log\left(\frac{n}{2}\right) \le T\left(\frac{n}{2}\right) \le c_2 \log\left(\frac{n}{2}\right)$

$$c_1 \log_{\frac{4}{3}} n - c_1 \log_{\frac{4}{3}} 2 \leq T\left(\frac{n}{2}\right) \leq c_2 \log n - c_2 \log 2 \leq c_2 \log n \mid + c$$

$$c_1 \log n - c_1 \log 2 + c \leq T(n) \leq c_2 \log n + c$$

c_1, c, c_2

$c_1 \log_2 + c \geq 0$

$c \geq c_1 \log 2$

$T(n) \in \theta(\log n)$

$T(n) = T\left(\frac{3n}{4}\right) + T\left(\frac{n}{4}\right) + n$

nivel 0: $T(n)$ ..... n

nivel 1: $T\left(\frac{n}{4}\right)$ $T\left(\frac{3n}{4}\right)$ ... $\frac{n}{4} + \frac{3n}{4} = n$

nivel 2: $T\left(\frac{n}{4^2}\right)$ $T\left(\frac{3n}{4^2}\right)$ $T\left(\frac{3n}{4^2}\right)$ $T\left(\frac{9n}{4^2}\right)$ .... $\frac{n}{4^2} + \frac{3n}{4^2} + \frac{3n}{4^2} + \frac{9n}{4^2} = n$

$\frac{1}{4}$ | ... $\frac{3}{4}$

S(n)

.....

h_1: $T\left(\frac{n}{4^{h_1}}\right)$ $T(\ldots)$ $T(\ldots)$

$S(n) \leq T(n) \leq G(n)$

$\theta(n \log n)$ $\theta(n \log n)$

.....

h_2: $T\left(\frac{3n}{4^{h_2}}\right)$

G(n)

h_1: $\frac{n}{4^{h_1}} = 1 \Rightarrow h_1 = \log_4 n$

h_2: $\frac{n}{\left(\frac{4}{3}\right)^{h_2}} = 1 \Rightarrow h_2 = \log_{\frac{4}{3}} n$

$$S(n) = \sum_{i=0}^{h_1-1} n + 2^{h_1} \cdot c = n\log_4 n + 2^{\log_4 n} \cdot c =$$

$$= n\log_4 n + n^{\log_4 2} \cdot c =$$

$$= n\log_4 n + \sqrt{n} \cdot c = \in \Theta(n \log n)$$

$$G(n) = \sum_{i=0}^{h_2-1} n + 1 \cdot c = h_2 \cdot n + c = \log_{\frac{4}{3}} n \cdot n + c = \Theta(n \log n)$$

$h_1 \rightarrow$ plin cu frunze

$h_2 \rightarrow$ 1 frunză

$h_2 - 1 \rightarrow$ plin de noduri

- $T(n) = \sqrt{n}\, T(\sqrt{n}) + n$

$n = 2^K \Rightarrow K = \log_2 n$

$T(2^K) = 2^{K/2} \cdot T(2^{K/2}) + 2^K \mid : 2^K$

$$\frac{T(2^K)}{2^K} = \frac{T(2^{K/2})}{2^{K/2}} + 1$$

$\frac{T(2^K)}{2^K} = M(K)$

$M(K) = M\left(\frac{K}{2}\right) + 1$

$a = 1 \quad c = \log_b a \Rightarrow M(K) = \Theta(K^0 \cdot \log K) = \Theta(\log K)$

$b = 2 \quad K = 0$

$c = 0$

$$\frac{T(2^K)}{2^K} = \log(K)$$

$$\frac{T(n)}{n} = \Theta(\log(\log n)) \qquad T(n) = \Theta(n \log(\log n))$$

$T(n) = 2T(n-1) + 1$

0: $T(n)$ 1

1: $T(n-1)$ $T(n-1)$ 2

2: $T(n-2)$ $T(n-2)$ $T(n-2)$ $T(n-2)$ 4

. . . . . .

k: $\boxed{T(n-k)}$ ..... ~~cu frunze~~ c · nr frunze

$T(1)$

$n - k = 1 \Rightarrow k = n-1$

$$T(n) = \sum_{i=0}^{k} 2^i + 2^k \cdot c = \frac{2^{k+1} - 1}{2-1} + 2^k \cdot c =$$

$$= 2^n - 1 + 2^{n+1} \cdot c = \Theta(2^n)$$

- $T(n) = 2T(\sqrt{n}) + n$

  # subst. că $T(n) \in \Theta(n)$

  Și că $T(\sqrt{n}) \in \Theta(\sqrt{n})$

  $\exists c_1, c_2 \in \mathbb{R}^*$: $c_1\sqrt{n} \leq T(\sqrt{n}) \leq c_2\sqrt{n} \mid \cdot 2$

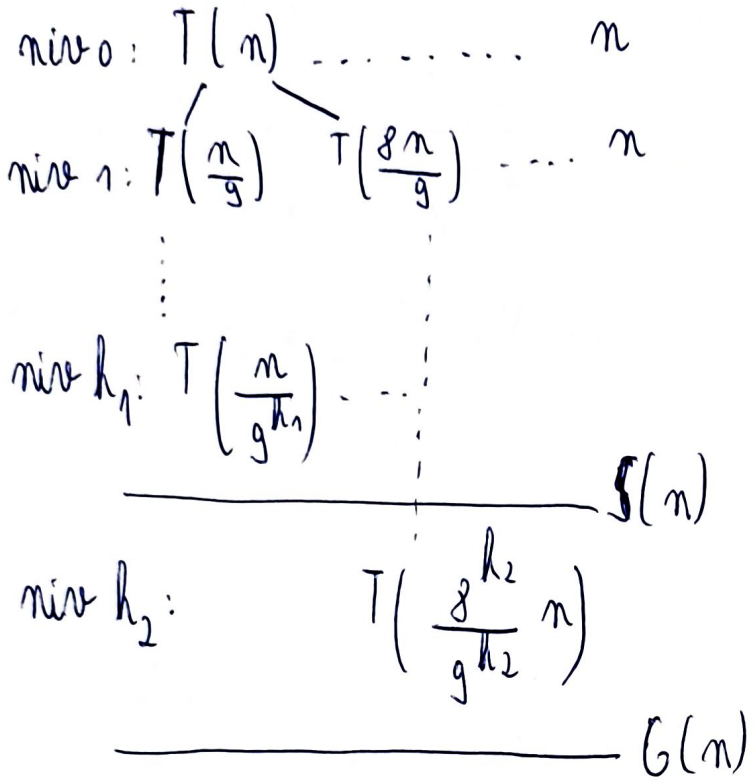
  $2c_1\sqrt{n} \leq 2T(\sqrt{n}) \leq 2c_2\sqrt{n} \mid + n$

  $n \leq 2c_1\sqrt{n} + n \leq 2T(\sqrt{n}) + n \leq 2c_2\sqrt{n} + n \leq 2n$

  $n \leq T(n) \leq 2n$

  $\Theta(n)$

$$T(n) = T\left(\frac{n}{9}\right) + T\left(\frac{8n}{9}\right) + n$$



$h_1 = \log_9 n$

$h_2 = \log_{\frac{8}{9}} n$

$$S(n) \leq T(n) \leq G(n)$$

$$S(n) = \sum_{i=0}^{h_1 - 1} n + 2^{h_1} \cdot c = n \log_9 n + 2^{\log_9 n} \cdot c = \Theta(n \log n)$$

$$n^{\log_9 2} < n \log n$$

$$G(n) = \sum_{i=0}^{h_2 - 1} n + 1 \cdot c = n \log_{\frac{9}{8}} n + c = \Theta(n \log n)$$

$$\Rightarrow T(n) = \Theta(n \log n)$$

Lab 8 - AA

SAT → verifică dacă o expresie logică este satisfiabilă

clauză$_1$ ∧ ... ∧ clauză$_n$ $\quad$ SAT ∈ NP

↳ literal$_1$ ∨ ... ∨ literal$_n$

Ex. $(x_1 \vee \bar{x}_2 \vee x_3) \wedge (x_1 \vee x_3) \wedge (\bar{x}_2 \vee x_1)$

input: "$(x_1 \vee x_2) \wedge x_3$"

$x_1 = 1$

$x_2 = 1$

$x_3 = 1$

output: 1

$(x_1 \vee x_2) \wedge (\bar{x}_2 \vee \bar{x}_1)$

$x_1 = 1$

$x_2 = 0$

⇒ output: 1

$x_1 \wedge \bar{x}_1$ ⇒ output: 0

$(\bar{x}_0 \vee \bar{x}_3 \vee x_5) \wedge x_3 \wedge (x_3 \vee x_6 \vee \bar{x}_4 \vee x_2 \vee x_1 \vee x_5) \wedge$

$\wedge (x_5 \vee \bar{x}_1 \vee \bar{x}_3 \vee x_0) \wedge (\bar{x}_1 \vee \bar{x}_5 \vee x_6)$

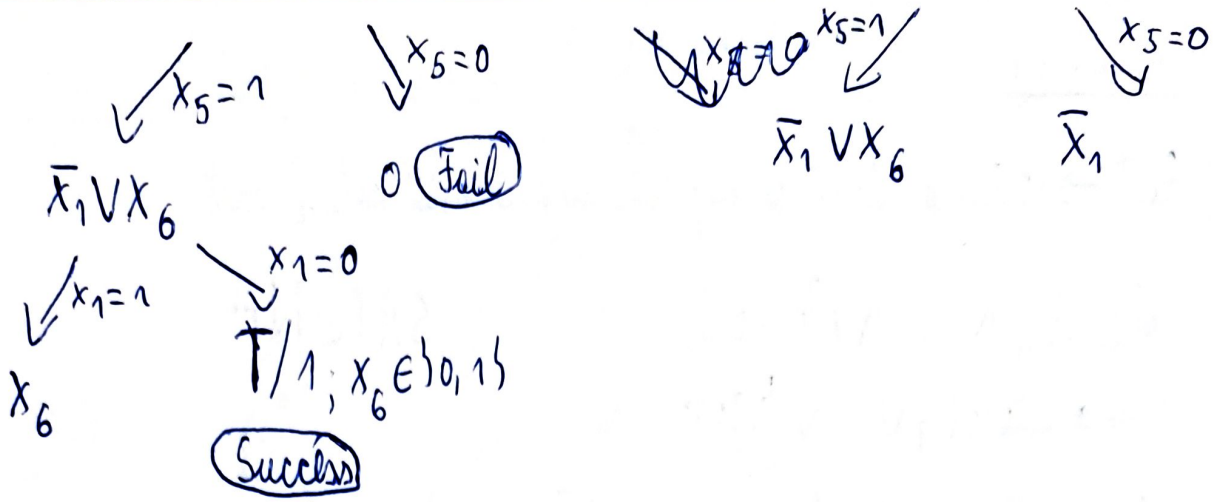
$x_3 = 1$ ↙ ↘ $x_3 = 0$

0 (fail)

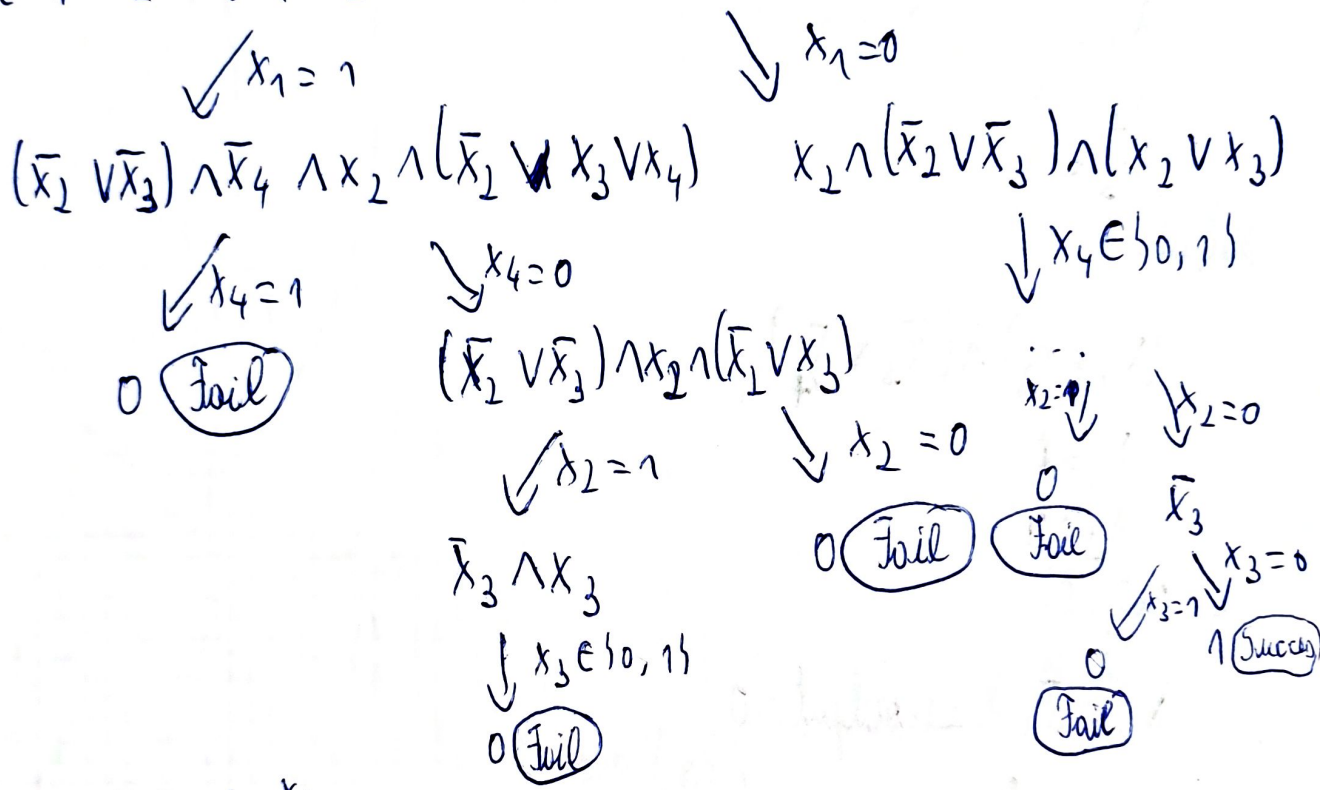
$(\bar{x}_0 \vee x_5) \wedge (x_5 \vee \bar{x}_1 \vee x_0) \wedge (\bar{x}_1 \vee \bar{x}_5 \vee x_6)$

$x_0 = 1$ ↙ ↘ $x_0 = 0$

$x_5 \wedge (\bar{x}_1 \vee \bar{x}_5 \vee x_6)$ $\qquad$ $(x_5 \vee \bar{x}_1) \wedge (\bar{x}_1 \vee \bar{x}_5 \vee x_6)$



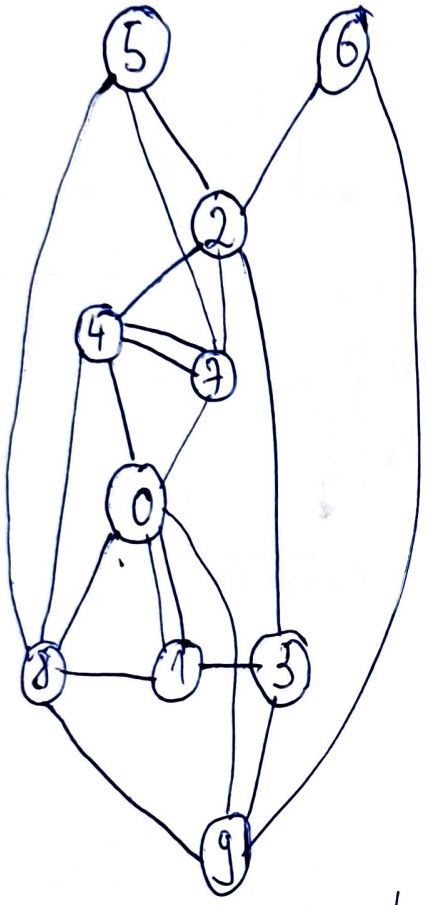
$$(x_1 \vee x_2) \wedge (\bar{x}_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee \bar{x}_4) \wedge (x_1 \vee x_2 \vee x_3) \wedge$$
$$\wedge (\bar{x}_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee x_5 \vee x_4)$$



$$I = \{0,1, 0,1\}$$
$$x_1\ x_2\ x_3\ x_4$$

Complexitate temporală angelică: $O(n)$

reală: $O(2^n)$



(K-Vertex Cover)

4 - Vertex Cover

[0, 1, 7, 2] FAIL

arce ↑ noduri ↑

KVC(G = (E, V); K)

real: $O(2^{|V|} \cdot |V|^2)$

cât branchuri

c=0 / c=1

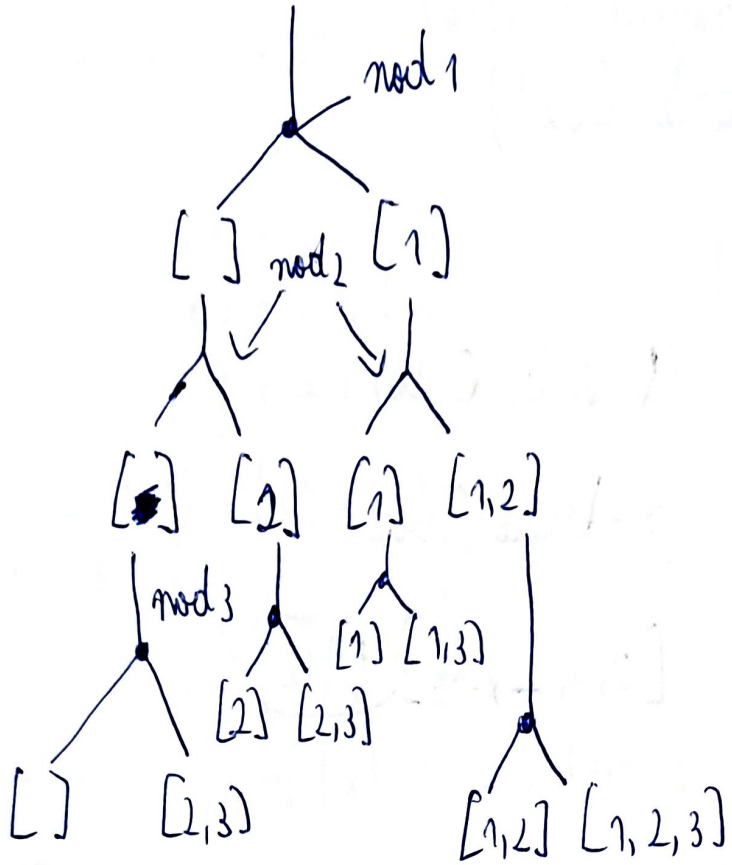
```
ales = []
for nod in V:
    c = choice({0, 1})
    if c==1: ales.append(nod)
if (len(ales) != K): fail
for (x,y) in E:
    if !(x in ales or y in ales): fail
success
```

generare de soluții $\{0(|V|)\}^{\infty}$

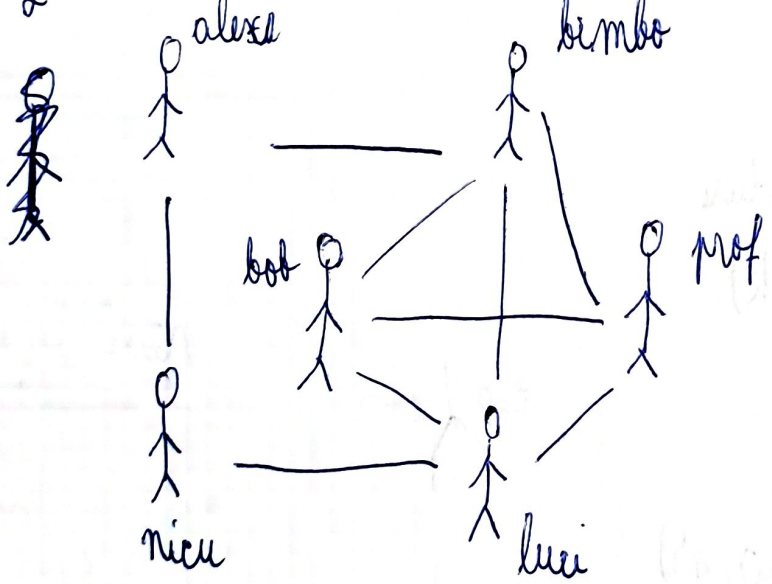
certificat polinomial $O(|V|^2)$

$O(|V|^2)$ angelic

! ← ! ← ! ← ! ∞ angelic



K Clique



K=4 : { bimbo, luci, prof, bob }

K=3 : { alexa, bimbo, bob }

K=5 ϕ

5 Clique = 0

4 Clique = 1

3

K Clique ($G = (E, V), K$):

reală: $O(2^{|V|} \cdot |V|^2) = O(2^n)$

gen. de sol angelică: $O(|V|)$

```
clig = []
for om in V:
    c = choice({0, 1}) ⟶ O(1)
    if c = 1: clig.append(om)
```

(F) certificat polinomial $O(|V|^2)$

```
if (len(clig) != K): fail
for x in clig:
    for y in clig:
        if (x != y and (x,y) ∉ E and (y,x) ∉ E): fail
success
```

$O(|V|^2)$

angelică

Lab 9 - AA

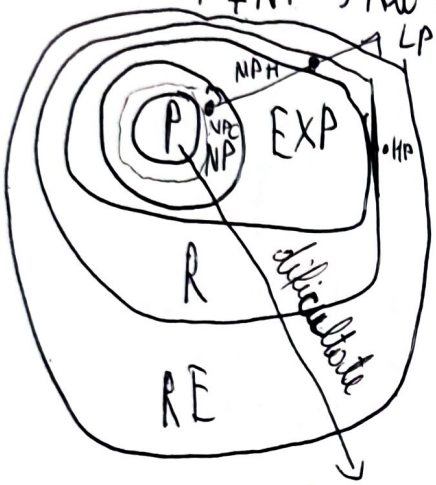
P = mulțimea de probleme ce pot fi decise în timp polinomial

NP = ——— // ——— nedeterministic polinomial

$P \subsetneq NP \subseteq EXP$

EXP = ——— // ——— exponențial

$P \neq NP$ → Nu e dem., dar se crede cu tărie



$f \leq \ldots \in R$ RE

$\ldots \notin R \leq f$

Reduceri polinomiale: $A \leq_p B$

A se reduce polinomial la B

B este mai grea ca A

$\exists$ o transformare T a.î.

$\forall w$

$A(w) = 1 <=> B(T(w)) = 1$

T se calculează în timp polinomial

$f \leq_p g \; / \Rightarrow g \notin P$
$f \notin P$

$f \leq_p g$
$g \in P \; / \Rightarrow f \in P$

R NP
RE EXP

$f, g \in NPC \quad f \leq_p g, g \leq_p f$

$NPH = \{ f \mid \forall y \in NP,\ y \leq_p f \}$

$NP\text{-}C = \{ f \mid f \in NP\text{-}H \text{ și } f \in NP \}$

NP ≠ P

$f \notin P$ → nu avem algoritm / nu putem dem.

$f \in NP\text{-}H$ – singură metodă care spune că $f \in P$

↳ $SAT \leq_p f$

$SAT \in NPC, NPH, \underline{NP}$

Ex: $(x_1 \vee \bar{x}_2) \wedge (x_3 \vee \bar{x}_2) \wedge (\bar{x}_4 \vee x_5 \vee x_3 \vee x_2)$

$I = \{ x_1 = T, x_2 = F, \dots \}$

$I \models \ell \iff \ell(I) = True$

- $f \in NPC$ / $y \in NPC$ | $f \leq_p y$? DA
  $y \leq_p f$? DA

$y \in NPH$

↳ $\forall x \in NP,\ x \leq_p y \Rightarrow f \leq_p y$

$f \in NPC \Rightarrow f \in NP$

$f \in NPH$

...

$y \in NP$

...

- $f \in NP$, $g \leq_P f$ $\Rightarrow$ $g \in NP$? — da
  $g \in P$ — nu (neaparat)

- $f \in P$, $g \leq_P f$ $\Rightarrow$ $g \in NP$ — da

  $\Downarrow$

  $g \in P$

- $f \in NPC$, $g \leq_P f$ $\Rightarrow$ ~~daca~~ $g \in NPC$? NU

  $\Rightarrow$ g poate fi și în P

  $g \leq_P \dots NP$

- $f \in NPC$ $\overset{?}{\Rightarrow}$ $g \in NPC$ NU
  $f \leq_P g$

  $\Rightarrow$ $f \in NP, NPH$

  $\dots NPH \leq_P g \Rightarrow g \in NPH$

  $g \leq_P \dots NP$ Nu avem relație

$f \leq_P SAT \Rightarrow f \in NPC$?

$SAT \in NPC$

↳ ~~NPH~~ $SAT \in NPH$

~~NP~~ $SAT \in NP$

$g \leq_P \dots NP \Rightarrow g \in NP$

$\dots NPH \leq_P g \Rightarrow g \in NPH$

$h \leq_p SAT$
$SAT \in NP \Big| \Rightarrow h \in NP$

Ca. $f \in NPC$, trebuie demonstrat că $f \in NP$ și $f \in NPH$
Pt. a $\# f \in NPH$, avem nevoie de o relație $h \leq_p f \in NPH$

- $f \leq_p g$
  $g \leq_p h \Big| \Rightarrow f \leq_p h$ [DA]
  $g \leq_T h$
  DA: $f \leq_T h$

Partitioning (A)

$A = [a_1, a_2, \ldots, a_n]$

$\exists S \subseteq A$ a.î. $\sum_{x \in S} x = \sum_{y \notin S} y$

Subset sum (B, K)

$B = [b_1, b_2, \ldots, b_m]$

$\exists S \leq B$ a.î. $\sum_{x \in S} x = K$

Partitioning $\leq_p$ Subset Sum

$A \xrightarrow[T]{} B, K$

T: $B = A$
$K = Sum(A)/2$
↳ $O(card(A))$ OK polinomială

„$\Rightarrow$" Part. (A) = True

$\exists S \subseteq A$ s.t. $\sum_{x \in S} x = \sum_{y \in A \setminus S} y = \frac{sum(A)}{2}$

$\Rightarrow \exists S \subseteq B$ s.t. $\sum_{x \in S'} x = K$

$\Rightarrow SS(B, K) = True$

"$\Leftarrow$" $SS(B, K) = True \Rightarrow \exists S' \subseteq B$ s.t. $\sum_{x \in S} x = K$

$\Rightarrow \exists S \subseteq A$ s.t. $\sum_{x \in S} x = \frac{sum(A)}{2} = \sum_{y \in A \setminus S} y \Rightarrow Part(A) = True$

- Subset Sum $\leq_p$ Partitioning

$B, K \xrightarrow[T]{} A$

$B = [1, 3, 5, \underbrace{2, 7}_{14}] \longrightarrow S = [1, 3] \cup \{10\}$

$K = 4$ $\quad Sum(B) - 2K$

$A = B \cup \{abs(Sum(B) - 2K)\}$

$2K < sum(B)$

"$\Rightarrow$" $SS(B, K) = True \Rightarrow \exists S \subseteq B$ s.t. $\sum_{x \in S} x = K$

$\Rightarrow \exists S' = S \cup \{sum(B) - 2K\}$ s.t. $\sum_{x \in S'} x = K + \underbrace{Sum(B) - 2K}_{\frac{Sum(A)}{2}} =$

$= Sum(B) - K$

$\Rightarrow Partitioning(A) = True$

"$\Leftarrow$" $Partitioning(A) = True$

$\Rightarrow \exists S \subseteq A$ s.t. $\sum_{x \in S} A = \sum_{y \in A \setminus S} y = \frac{Sum(A)}{2}$

Alegem S a.î. să aibă elem. $(sum(B)-2K)$

$\Rightarrow \sum_{y \in S \setminus \{sum(B)-2K\}} y = \frac{sum(A)}{2} - sum(B) - 2K =$

$= sum(B) - K - (sum(B) - 2K) = K$

$\exists S' \subseteq B$ a.î. $\sum_{y \in S'} y = K \Rightarrow SS(B, K) = True$